SIMPLEST FORM

Reducing the index

Converting to a rational exponent makes the reduction mechanical: write the exponent as a fraction, reduce it, convert back.

01 RADICAL, INDEX 6 — BEFORE

$$\sqrt[6]{x^4}$$

↓ Write as a rational exponent — the index becomes the denominator

02 EXPONENT FORM

$$x^{4/6}$$

↓ Reduce the fraction — 4 and 6 share a common factor of 2

03 REDUCED EXPONENT

$$x^{2/3}$$

↓ Convert back — the denominator 3 is the new index

04 RADICAL, INDEX 3 — AFTER

$$\sqrt[3]{x^2}$$